

# Ethan Lee

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## EDUCATION

### University of California, Los Angeles

*Bachelor of Science in Computer Science*

Los Angeles, CA

*Sep. 2023 – June 2027*

- **Relevant Coursework:** Data Structures & Algorithms, Software Engineering, Computer Organization, Software Construction, Machine Learning, Operating Systems, Computer Graphics, Fundamentals of AI, Discrete Math

## EXPERIENCE

### Capital One

*Software Engineer Intern*

Richmond, VA

*Jun. 2026 – Aug. 2026*

- Architected 3-tier serverless Go API on AWS Lambda adhering to OpenAPI 3.0, paired with an Angular dashboard.
- Wrote Go unit tests, configured a Jenkins CI pipeline, and monitored deployments via CloudWatch queries.
- Streamlined alert evaluation w/ EventBridge and CloudWatch, eliminating 5 hrs/week of manual cost auditing.
- Integrated the alerting feature into Capital One's cloud platform (5+ engineer adopters); presented to the Cloud org.

## PROJECTS

### Reflectory | *Next.js, TypeScript, Supabase/Postgres, pgvector, Gemini API*

Sep. 2026 – Present

- Built an AI journaling companion grounding reflections in a user's entry history via RAG (pgvector + Gemini).
- Designed a two-layer crisis-detection system (keyword + LLM classifier) to reroute at-risk users to crisis resources.
- Enforced per-user data isolation via Supabase RLS, verified through independent multi-account testing.
- Implemented atomic Postgres RPC rate-limiting across Vercel's multi-instance runtime, backed by 112 Vitest tests.

### The Primer | *React, TypeScript, Python, FastAPI, PostgreSQL, Docker, Gemini API*

Aug. 2026 – Present

- Built an adaptive math tutor for kids via Bayesian Knowledge Tracing, capped to 1 difficulty level/attempt.
- Decoupled deterministic grading from an LLM explanation layer so a Gemini outage degrades gracefully.
- Optimized PostgreSQL access with targeted indexes and eager-loading to eliminate an N+1 query pattern.
- Built a React frontend with a JWT-authenticated parent dashboard tracking per-skill mastery and session history.

### TrialFlow | *React, TypeScript, Express.js, Creao, Composio MCP, Google APIs*

Oct. 2025

- Won Regeneron Runner-Up at CalHacks 12.0 (600+ teams) tackling clinical trials' ~40% dropout rate.
- Constructed a custom Burden Score API via Creao to quantify participant workload based on procedure complexity.
- Automated patient self-scheduling by integrating Google Calendar APIs for real-time appointment sync.
- Orchestrated 24-hour notification pipelines delivering appointment logistics and personalized prep instructions.

### Bruin Plan | *MongoDB, Express.js, React, Node.js, Google OAuth*

Mar. 2025 – May 2025

- Architected MERN stack app processing 15,000+ course combinations to optimize student schedules.
- Built a UCLA course-catalog scraper enabling students to sort schedules by proximity or time of day.
- Integrated Google OAuth 2.0 authentication, improving signup-to-login conversion from 55% to 80%.
- Developed dynamic programming algorithm reducing average schedule conflicts from 3.1 per user to 2.0.

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Go, Python, C++, HTML/CSS, SQL (PostgreSQL, MongoDB)

**Frameworks:** React, Next.js, Angular, Express.js, FastAPI, Node.js, Tailwind CSS, SQLAlchemy

**Tools & Cloud:** AWS (Lambda/RDS/EventBridge), GCP, Docker, Supabase, Vercel, Render, Git, Linux

**AI & ML:** Gemini API, RAG (pgvector), Prompt Engineering, Bayesian Knowledge Tracing